Using Peridynamics to Model Hydrogen Embrittlement in Steel Pipes

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Abstract

Hydrogen embrittlement (HE) causes steel pipes to fracture at lower strains and with less warning than unexposed steel, yet experimental data on HE are limited and fracture models often require the crack path to be specified in advance. We pose the matching of experimental tensile stress–strain curves as an inverse problem for a state-based peridynamic correspondence model with Voce hardening and Brozzo ductile damage, and use it to learn material properties of TBF1200 steel at three hydrogen concentrations. The parameters are calibrated with Dakota in two stages, hardening first and damage second. The calibrated model reproduces the experimental fracture strains to within 0.5 percentage points. Embrittlement is captured almost entirely by a single parameter governing the onset of bond weakening, which falls from 0.85 to 0.19 as hydrogen concentration increases, while the damage-rate parameters are unchanged. We then apply the calibrated parameters, unmodified, to a pipe fragmenting under internal pressure, with no prescribed fracture path. As hydrogen concentration increases, failure begins earlier and at lower plastic strain, with the mean plastic strain at failure decreasing from 0.214 to 0.141, and the pipe absorbs less energy through plastic deformation before it fragments. These results show that peridynamic inverse calibration can extend limited tensile data to predict how hydrogen affects the failure of a pipe.

Keywords: Peridynamics, Hydrogen embrittlement, Steel pipes, Fracture, Calibration

1 Introduction

Hydrogen embrittlement (HE) is the loss of ductility and toughness that occurs when hydrogen enters a metal. It is a concern for high-strength steels in service, and most often it is encountered in pipes and pressure vessels, so predicting how a pressurized steel pipe fails as its hydrogen content increases is of practical interest [1]. At the microstructural level, hydrogen penetrating the steel creates voids; under load, these voids act as stress concentrations that nucleate cracks [2]. At the structural scale, the consequence is a transition from stable, ductile deformation to early and abrupt fracture.

Experimental data on steel with different hydrogen concentrations make this transition visible in the tensile response. In the tests of Nieto et al. [3] on TBF1200 sheet steel, which we use here, the stress–strain curve of the uncharged steel softens visibly over several percent strain before fracture. As the hydrogen concentration increases, this softening disappears and the material stays near its peak stress until it fractures abruptly at a much lower strain. Capturing this change in behavior, rather than only the drop in fracture strain, is a stringent test of any model of HE. Experimental data of this kind are limited, however. Only a few hydrogen concentrations have been tested, and they cannot cover the range of conditions in which HE matters in practice, including the geometry and loading of a pipe.

The harder question is how to use such limited, material-level data to predict HE in a pipe. Fracture models that represent cracks explicitly typically require the crack path to be specified, or a crack to be seeded, before the simulation begins. For a pressurized pipe this is a serious limitation, because the fracture pattern is itself the result we are after and it is not known in advance. Phase-field models avoid this requirement and have been applied to hydrogen-assisted cracking [4, 5], but they introduce an additional field that must be solved for alongside the displacement. Peridynamics [6–8] offers another alternative. It is a nonlocal reformulation of continuum mechanics in which each point interacts with its neighbors within a finite horizon through integral equations, rather than through spatial derivatives. Damage is represented by bond failure, so cracks can nucleate, branch, and coalesce without a prescribed path and without additional fields to track the crack. These properties make peridynamics well suited to fracture and fragmentation problems like the one considered here.

Peridynamics has previously been applied to hydrogen embrittlement, primarily by modeling the diffusion of hydrogen through a material and the cracking that results from the embrittlement it causes [9–11]. For example, Chen et al. [11] use a two-parameter bond-based model with a hydrogen-dependent bond-failure criterion. Such

a simplified bond-level constitutive description may limit the representation of plastic hardening and ductile damage that can be important in structural failure. Our approach addresses a different problem: we use a state-based correspondence model with an elastoplastic hardening law and ductile damage, calibrated to experimental stress–strain data at each hydrogen condition. We do not model hydrogen transport; instead, we use the calibrated material properties to simulate pipe failure.

These limited data motivate the approach taken here. Rather than using the stress–strain curves only to describe the specimens that were tested, we use them as ground truth for an inverse problem within a peridynamic framework: we seek the parameters of a state-based peridynamic correspondence material model, which allows a standard local constitutive law (here, Voce hardening) to be used within peridynamics, together with the Brozzo ductile damage law [12], that reproduce the measured tensile response of Nieto et al. [3] at each hydrogen concentration. The inverse problem is solved with Dakota [13]. Learning the material properties in this way extends the reach of the limited HE data. The calibrated parameters describe the material, not a single test, so they can be used to simulate configurations that were not tested and to assess the performance of pipes that have been exposed to hydrogen. We demonstrate this by applying the calibrated parameters, unmodified, to a fragmenting pipe loaded by internal pressure, and examining how the onset of fracture, the plastic strain at failure, and the distribution of damage change with hydrogen concentration. The main contributions are:

- a two-stage calibration procedure that separates hardening from damage, and shows that the Brozzo damage rates transfer unchanged between hydrogen concentrations while the bond-weakening onset parameter tracks embrittlement;
- a demonstration that parameters learned from a limited set of tensile tests can be used, without recalibration, to simulate a fragmenting pipe in which the fracture path is not prescribed; and
- a quantitative comparison of how pipe failure changes with hydrogen concentration, in terms of failure time, plastic strain at failure, and accumulated damage.

The remainder of the paper is organized as follows. Section 2 describes the peridynamic model, the hydrogen concentration cases, the calibration procedure, and the pipe simulation. Section 3 presents the calibration results and the pipe simulations, and Section 4 concludes.

2 Methods

Material parameters governing hardening and ductile damage were calibrated against experimental engineering stress–strain curves from slow-rate dogbone tensile tests at several hydrogen concentrations, using peridynamic simulations of the tensile test posed as an inverse problem and solved with Dakota. The calibrated parameters

for each concentration were then used, unmodified, to simulate a pressurized cylinder and study how the resulting damage and fragmentation behavior changes with embrittlement level.

2.1 Peridynamic formulation

Simulations were performed in Peridigm [14] using a state-based peridynamic [8] hypoelastic correspondence material model with isotropic hardening, in which the local deformation gradient is reconstructed from the relative motion of points within a finite horizon and used to evaluate a standard (local-form) constitutive response. The numerical formulation and implementation follow the semi-Lagrangian constitutive correspondence framework described by Behzadinasab and Foster [15]. Hardening followed a Voce-type saturation law [16], augmented by a linear hardening term,

$$\sigma_y(\varepsilon_p) = \sigma_0 + (\sigma_{\text{sat}} - \sigma_0)\left(1 - e^{-n\varepsilon_p}\right) + H\varepsilon_p, \tag{1}$$

where ε_p is the equivalent plastic strain and σ_0 (Initial Yield Stress), σ_{sat} (Saturated Yield Stress), n (Exponential Constant) and H (Linear Constant) are calibrated. H is a plastic hardening modulus, active only after yield, and is distinct from the material's elastic moduli (bulk modulus 130 GPa, shear modulus 78 GPa), which were fixed at values representative of the base TBF1200 steel and not calibrated. The corresponding Young's modulus, $E = 9KG/(3K + G) \approx 195$ GPa, agrees with the measured value of 201 GPa reported for this steel [3] to within 3%.

Ductile damage was modeled with the Brozzo/Wellman damage law of Behzadinasab and Foster [12], in which bond-level damage accumulates at a rate

$$\dot{D} = d_1 \left\langle \frac{2\sigma_T}{3(\sigma_T - \sigma_M)} \right\rangle^{d_2} \dot{\bar{\varepsilon}}_p, \tag{2}$$

driven by the bond-level maximum principal stress σ_T and hydrostatic stress σ_M, and Heaviside-gated to tension ($\langle\cdot\rangle$) so that damage accumulates only under tensile triaxiality. A bond retains its full force contribution until its accumulated damage D exceeds the onset value D_{onset}. Beyond this value, its force contribution decreases linearly with further damage accumulation, reaching zero (full bond failure) at $D = 1$. Point-level damage, reported in the Results, is the neighborhood-volume-weighted average of this bond-level quantity over a point's horizon. The horizon used for the dogbone specimen was 0.236 cm for neighbor search, with an actual horizon of 0.189 cm used internally by the correspondence material's gradient/shape-tensor calculation, set empirically rather than from a fixed rule.

2.2 Hydrogen concentration cases

The tensile stress-strain data used to calibrate the model, for all three hydrogen concentrations, were taken from the experimental work of Nieto et al. [3], who charged TBF1200 sheet steel specimens with hydrogen and measured the resulting tensile response. Three cases were studied, denoted HE_0, HE_{30}, and HE_{60}, where the subscript is the duration, in minutes, of electrolytic hydrogen charging (condition A30) applied to the TBF1200 sheet steel specimen before testing (HE_0 being an uncharged baseline). Measured bulk hydrogen concentration increases monotonically with charging time: 0, 0.68 and 1.16 mppm for HE_0, HE_{30}, and HE_{60} respectively [3]. Experimental fracture strain decreases monotonically with increasing concentration (15.37%, 13.27% and 7.68% for HE_0, HE_{30}, and HE_{60} respectively), and the character of the pre-fracture curve changes along with it: at low concentration the material softens visibly over several percent strain before fracturing, while at high concentration the material remains near peak stress until a comparatively abrupt final failure.

2.3 Calibration against tensile data

The overall calibration approach follows Behzadinasab and Foster's revisiting of the third Sandia Fracture Challenge [12]: material parameters are identified from calibration tests and then applied to a different geometry. Here, the calibration is adapted to tensile data at each hydrogen condition, using the two-stage procedure described below.

The dogbone specimen was modeled with a prescribed-displacement boundary condition at each grip, scaled by each node's own distance from the specimen center so that grip motion is affine rather than rigid: each grip node's y-displacement was prescribed as $u_y = y \cdot \dot{e} \cdot t$, with elongation-rate constant $\dot{e} = 45.0\ \mathrm{s}^{-1}$, common to all concentrations, run to a final time of 1.84 ms. At the grip node sets (located at $y \in [4.70, 5.08]$ cm and $y \in [-5.08, -4.70]$ cm), this gives nominal grip velocities of roughly 211–229 cm/s. The reported engineering stress and strain were not read directly from this boundary displacement, but from two tracked points at $y = \pm 1.238$ cm (gauge length 2.476 cm): strain is their relative displacement divided by the gauge length, and stress is the grip reaction force divided by the original gauge cross-section ($0.635 \times 0.315\ \mathrm{cm}^2 = 0.200\ \mathrm{cm}^2$), so that the localized necking within the gauge section is reflected rather than averaged out over the whole specimen. A finite horizon (as above) was used throughout. Simulations were explicit-dynamics, integrated with Peridigm's Verlet time-stepper, and solved in parallel across four MPI ranks. Although the tensile experiments were conducted quasi-statically, the simulations were loaded much more rapidly to reduce the computational cost of explicit time integration during calibration. The hardening and damage laws do not include an explicit strain-rate dependence, so this acceleration does not directly change the constitutive response at a given deformation state. It can, however, introduce inertial effects and small dynamic oscillations in the simulated response; the oscillatory noise

was reduced by smoothing the simulated stress–strain curves before calibration, as described below.

Calibration was driven by Dakota [13], an open-source toolkit developed at Sandia National Laboratories that couples an external simulation code to a library of optimization, parameter-estimation, uncertainty-quantification and sensitivity-analysis methods. Dakota treats the simulation as a black box: it proposes a set of parameter values, writes them to an input file, launches the forward simulation (here, the Peridigm dogbone model), reads back the resulting response, and uses it to propose the next set of parameters. In this way Dakota acts as the driver of the optimization loop, while Peridigm supplies the forward solve at each iteration. Specifically, we used Dakota's `coliny_pattern_search` algorithm, a local, derivative-free direct search. The objective function was defined as the mean relative error between simulated and experimental stress. Simulated stress–strain curves were interpolated at the experimental strain data points to evaluate the mean relative error. Prior to the comparison, smoothing was applied to the simulated curve to reduce noise resulting from the dynamic loading nature of the simulation. Experimental points below 1% of the curve's peak stress were excluded, since near-zero-stress sensor noise would otherwise make the relative error unstable. Where damage was active, a fracture-strain term (the relative error between simulated and experimental fracture strain) was added to the stress term, and the stress comparison was restricted to strains at or below the simulated fracture strain, since the two halves of a fractured specimen no longer produce physically meaningful stress readings. Convergence at each stage was assessed by confirming that perturbing each active parameter away from the current best point, in multiple directions, consistently failed to further reduce the objective. The resulting fit was also inspected qualitatively against the experimental curve before advancing to the next stage.

Calibration was performed in two stages. In Stage 1, only the hardening parameters (σ_0, σ_{sat}, n, H) were treated as free. Damage was fixed so that it never triggered, using a high bond-weakening onset value and a damage-rate constant well above any plastic strain reached in this stage. The fit therefore reflected hardening alone. The fracture-strain term was also dropped from the objective at this stage.

For every concentration after HE_0, Stage 1 was warm-started from the nearest calibrated concentration. We expected hardening to transfer well across embrittlement level, and this was confirmed for all three concentrations. Initial Yield Stress was the only Stage 1 parameter that moved meaningfully across concentrations. Saturated Yield Stress, the Exponential Constant and the Linear Constant were essentially unchanged.

In Stage 2, the hardening parameters were held fixed while the Brozzo damage parameters were calibrated. The onset parameter D_{onset} was swept with d_1 and d_2 fixed; for concentrations after HE_0, their values were transferred from the previously calibrated concentration. Across concentrations, d_1 and d_2 remained essentially unchanged, whereas D_{onset} varied with embrittlement level. For HE_{30}, a subsequent

joint search over all three damage parameters yielded a modest improvement, and those values are reported. For HE_0, d_1 and d_2 were jointly optimized with D_{onset} fixed. For HE_{60}, the reported values come from the onset sweep because the joint damage-parameter search did not converge. Hardening and damage parameters were not jointly re-optimized for any concentration.

2.4 Fragmenting pipe

The pressurized cylinder was discretized with four points through the wall thickness and a 10% random perturbation of node positions to break mesh symmetry and let fracture localize naturally, with a horizon of 4.17 mm. The same random seed was used for all three hydrogen concentrations, so the perturbed point positions are identical across cases. The cylinder was loaded with an internal pressure, applied as a radially directed body force on the inner surface node layer and ramped linearly from 0 to 500 MPa over the simulation's 400 μs duration, together with a matching body force on the end-cap node sets representing the axial stress a closed-end pressure vessel develops under that same internal pressure. Each concentration's calibrated hardening and damage parameters (Table 1) were applied unmodified; no parameters were recalibrated against the cylinder geometry. Fragments were identified from the resulting bond connectivity at the end of the simulation.

3 Results

3.1 Dogbone calibration

Calibrated parameters for the three hydrogen concentrations are listed in Table 1, and the resulting simulated stress–strain curves are compared with the experimental data in Fig. 1. In Stage 1, hardening was nearly independent of hydrogen concentration. Only the initial yield stress changed: for HE_{30} it was 1,027 MPa, about 5% higher than the 977 MPa found for HE_0 and HE_{60}. In Stage 2, embrittlement was captured almost entirely by the bond-weakening onset value D_{onset}, with d_1 and d_2 unchanged. D_{onset} falls by 40% from HE_0 to HE_{30} (0.85 to 0.51) and is 0.19 for HE_{60}, less than a quarter of the HE_0 value. This reduction causes the HE_{60} specimen to fracture at about half the strain of HE_0 (7.80% vs. 15.87%). Figure 2 shows that the simulated fracture strain decreases with hydrogen concentration and agrees with experiment to within 0.5 percentage points for all three cases, which demonstrates the embrittlement captured by the model. These parameters were then applied, unmodified, to the pressurized cylinder.

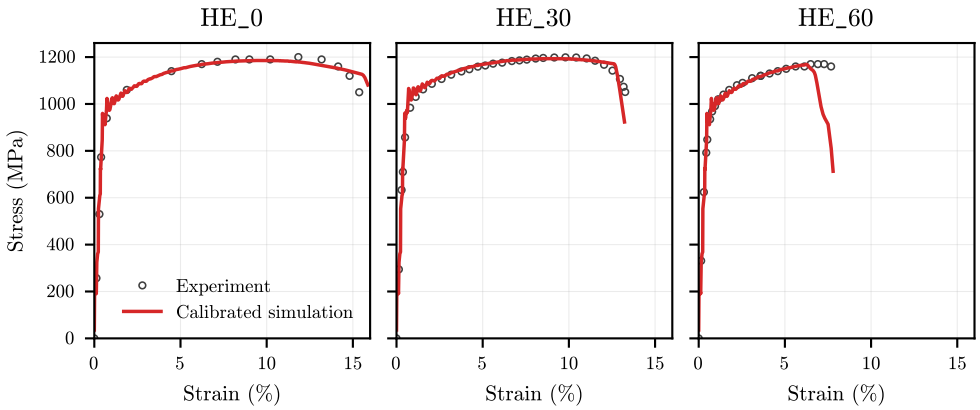


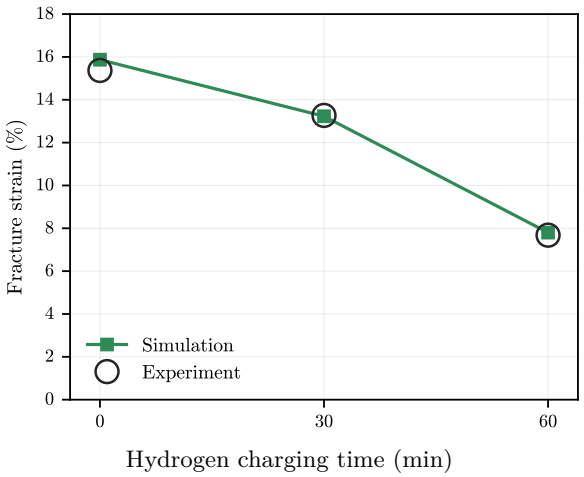
Fig. 1: Experimental (markers) and calibrated simulated (line) engineering stress–strain curves for HE_0, HE_{30} and HE_{60} at the final calibrated parameters (Table 1). The simulated curves are smoothed with a Savitzky–Golay filter to remove loading noise up to 10% strain; from 10% strain to the simulated fracture strain the unfiltered curves are shown. All panels share the same strain axis.

Table 1: Calibrated material parameters for the three hydrogen concentrations. Stage 1 fits the Voce hardening law (yield and plastic flow) to the tensile curve with damage disabled; Stage 2 then calibrates the Brozzo damage model (post-yield, through fracture) with the Stage 1 hardening parameters frozen. Density (7800 kg/m^3), bulk modulus (130 GPa) and shear modulus (78 GPa) are identical for all three cases and were not calibrated.

	HE_0	HE_{30}	HE_{60}
Stage 1: hardening (Voce), damage disabled			
Initial yield stress (MPa)	977	1027	977
Saturated yield stress (MPa)	1350	1350	1350
Exponential constant	22.0	22.0	22.0
Linear constant (MPa)	200	200	200
Stage 2: damage (Brozzo), hardening frozen			
d_1	3.5	3.5	3.5
d_2	0.99	0.99	0.99
D_{onset}	0.85	0.51	0.19

3.2 Fragmenting pipe

Figure 3 shows damage in the pressurized cylinders at three times for each hydrogen concentration. The fragmentation pattern is nearly identical across the three cylinders because the same heterogeneity, added to mimic the heterogeneous microstructure of metals, is imposed in each case. Without it, cracks would not nucleate naturally. The



Fracture strain (%)		
Case	Exp.	Sim.
HE_0	15.37	15.87
HE_{30}	13.27	13.23
HE_{60}	7.68	7.80

Fig. 2: Fracture strain as a function of hydrogen charging time. In the plot (left), the simulated fracture strain from the calibrated model (squares and line) is compared with the experimental fracture strain (open circles) for HE_0, HE_{30} and HE_{60}. The table (right) lists the same values. Fracture strain decreases with increasing hydrogen concentration in both the experiments and the simulations.

differences between concentrations therefore lie in when cracks initiate and in how much damage accumulates within the fragments.

These snapshots are shown at later times (up to 386.7 μs) than the quantitative damage statistics in Figs. 4 and 5 (capped at 331 μs), since observing the fragmentation process itself, rather than only the damage state leading up to it, requires times past initial wall breach. The applied pressure continues to ramp on its prescribed schedule even after the wall breaches, with no venting or depressurization coupling, so the loading magnitude at these later times likely runs higher than a real depressurizing pipe would experience. This does not affect the qualitative crack-initiation-timing comparison between concentrations shown here, which is established by material behavior before breach, but is why quantitative statistics elsewhere are only reported up to 331 μs.

At the earliest time shown ($t = 380.8$ μs, top row), only the HE_{60} cylinder contains points with damage above 0.5 (warm colors), and these are spread along its length. Cracks initiate at these locations, as seen in the growing gaps between fragments in the subsequent frames. In the HE_{30} cylinder, fragmentation first appears at the second time ($t = 384.7$ μs), with gaps forming between fragments by the third. In the HE_0 cylinder, cracks are clearly visible only at the third time ($t = 386.7$ μs), and the gaps between fragments are much smaller than in the other two. The HE_0 steel can strain plastically and absorb energy, which delays fracture and fragmentation, whereas the embrittled HE_{60} steel fractures at lower plastic strain and therefore fragments sooner.

The damage within the fragments also differs between concentrations. Fragments of the HE_{60} cylinder are mostly blue, indicating little accumulated damage, whereas fragments of the HE_0 cylinder are largely warm-colored. The uncharged HE_0 steel

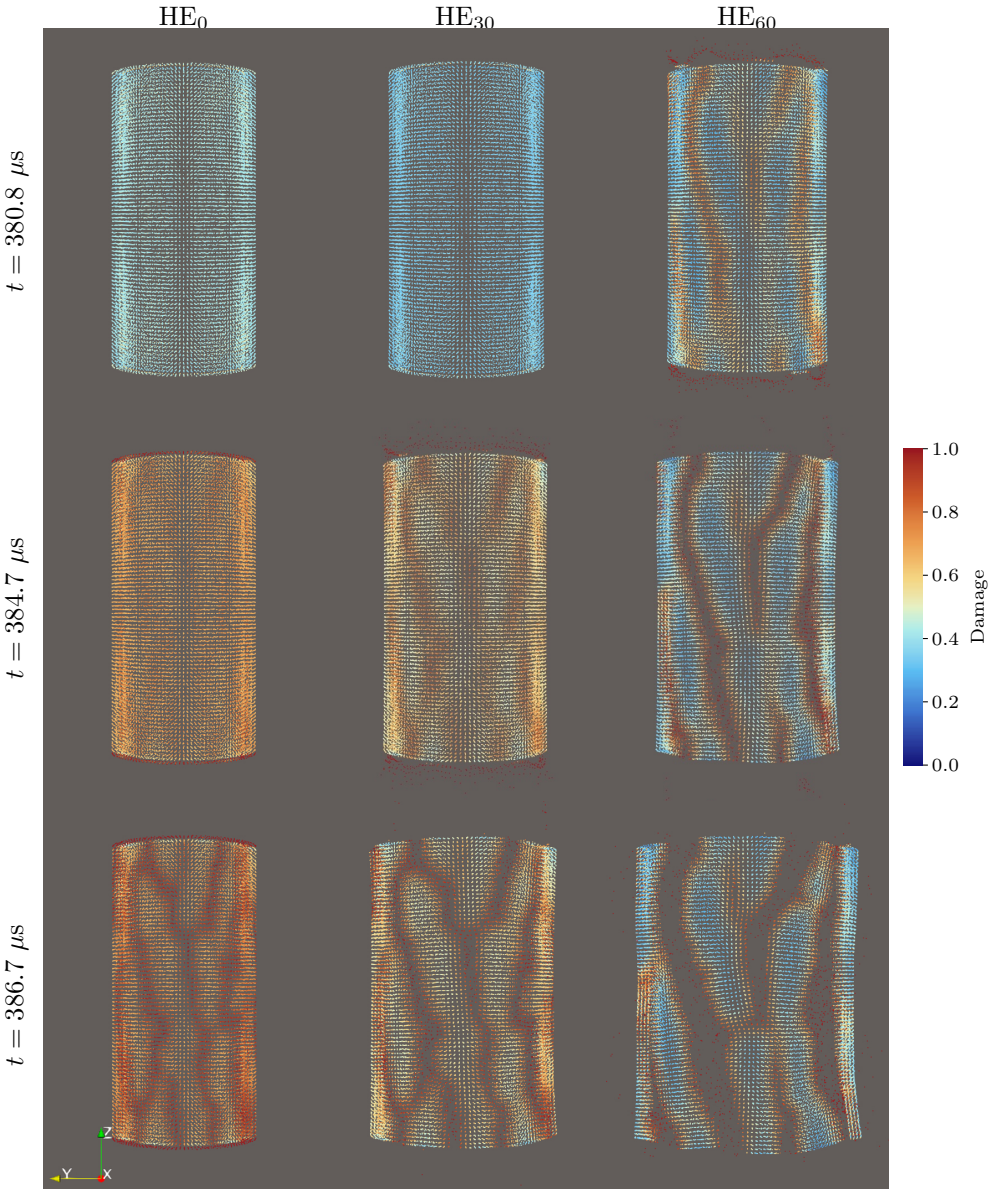


Fig. 3: Damage in the fragmenting cylinder at $t = 380.8$, 384.7 and 386.7 μs (top to bottom) for HE_0, HE_{30} and HE_{60} (left to right). Cracks open earlier in time as hydrogen concentration increases; the colorbar gives point damage.

strains plastically as it is loaded, which spreads the load over more points and raises their damage, but this plastic deformation also absorbs energy that would otherwise drive fracture. As a result, HE_0 contains many points with damage above 0.5 and yet fractures later than the embrittled cylinders.

The plastic strain accumulated by each point at the moment it fails quantifies this trend. As shown by the cumulative distributions in Fig. 4, points in the more heavily charged cylinders fail at progressively lower plastic strain. This is reflected in the mean plastic strain at failure, which decreases from 0.214 for HE_0 to 0.176 for HE_{30} and 0.141 for HE_{60}. Figure 5 shows the distribution of point damage at 331 μs. In HE_{60}, 49% of points have a damage value of 0.49 or less, compared with none in HE_0. Together, these results show that as hydrogen concentration increases, the pipe's capacity to deform plastically and absorb energy decreases, so fracture initiates earlier in time and at lower plastic strain.

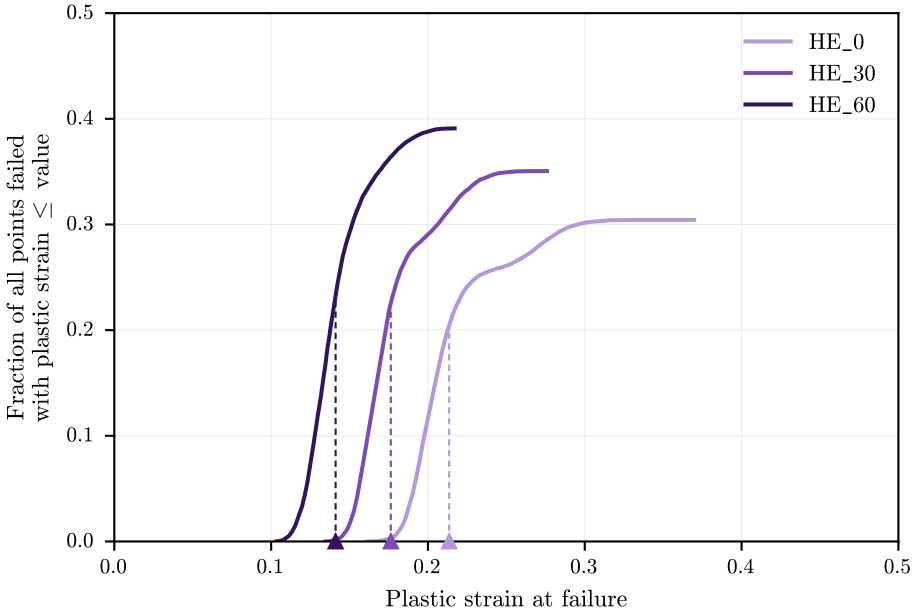


Fig. 4: Plastic strain at failure in the pressurized pipe for HE_0, HE_{30} and HE_{60}. For each point that failed (more than half of its bonds broken), we record the plastic strain it had accumulated at that moment. Each curve shows the fraction of all points in the pipe that failed with plastic strain at or below the value on the horizontal axis. Failures are counted up to 331 μs, so the curves plateau at the fraction of points that had failed by then: 30%, 35% and 39% of points for HE_0, HE_{30} and HE_{60}, respectively. The median plastic strain at failure is 0.205 for HE_0, 0.170 for HE_{30} and 0.137 for HE_{60}. Triangles on the horizontal axis, with dashed lines up to each curve, mark the mean plastic strain at failure of the points that had failed, which is 0.214, 0.176 and 0.141 for HE_0, HE_{30} and HE_{60}, respectively.

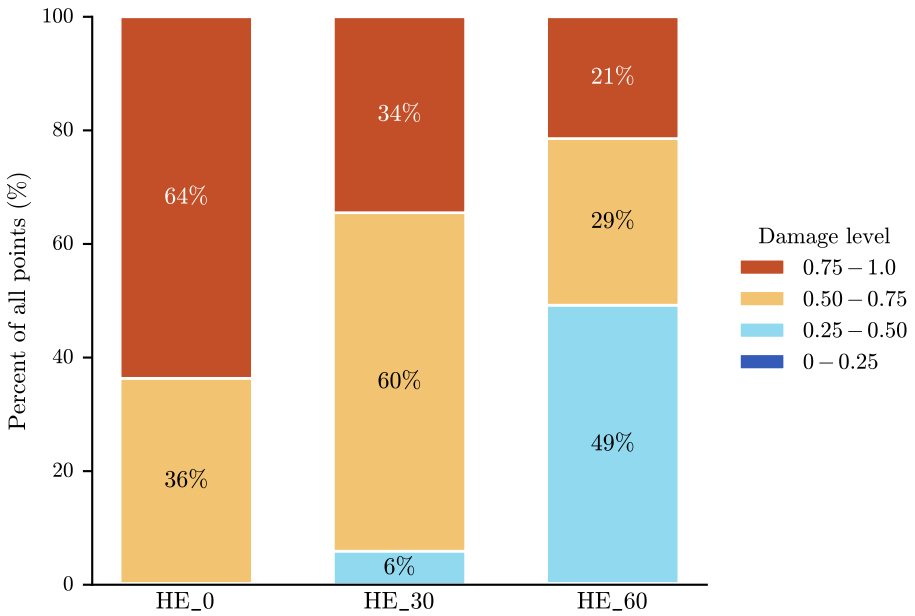


Fig. 5: Share of material points in each accumulated-damage class for the pressurized pipe with HE_0, HE_{30} and HE_{60} at $t = 331$ μs (internal pressure ≈ 414 MPa, from a linear ramp of 0 to 500 MPa over 400 μs). Each bar represents 100% of the points in the cylinder, split into four damage classes whose colors follow the damage colorbar in Fig. 3. Points with damage above 0.75 make up 64%, 34% and 21% of the cylinder for HE_0, HE_{30} and HE_{60}, respectively.

4 Conclusion

We used a state-based peridynamic correspondence model to learn the properties of TBF1200 steel at three hydrogen concentrations from experimental tensile curves, and then used those properties to simulate a pipe fragmenting under internal pressure. The calibrated model reproduced the experimental fracture strains to within 0.5 percentage points, and embrittlement was captured almost entirely by the bond-weakening onset value D_{onset}, which decreased from 0.85 to 0.51 to 0.19 while d_1 and d_2 did not change. The results demonstrate that the model can effectively capture the effects of hydrogen on the mechanical behavior of the material. Applied unmodified to the pipe, with no prescribed crack path, the calibrated parameters produced earlier failure at lower plastic strain as hydrogen concentration increased, with the mean plastic strain at failure decreasing from 0.214 to 0.141. The inverse problem thus extends a small set of tensile tests to a configuration that was not tested.

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